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香 港 大 學
THE UNIVERSITY OF HONG KONG

CIVL2108 Principles of Civil Engineering Management

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Construction Planning

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Table 1.5 The planning process of the construction client.

Project plan	Action
1. During the design stage	Appraise options
	Confirm business case
	Develop project strategy
	Prepare strategic brief
	Assemble team
	Devise risk management plan
	Choose procurement arrangement
	Risk assessments and pre-construction information
	Health and safety plan
	Commence health and safety file
	Prepare client programme (master schedule)
	Budget and cash flow
	Pre-qualify contractors
	Organise and administer tender stage
	Check tenders
	Choose preferred bid
	Prepare contract documents
	Sign contract
2. During the construction stage	Pre-start meeting
	Check bonds and Insurances
	Check construction health and safety Plan
	Contract administration
	Make contractor payments
	Monitor progress
	Report to the client
	Handover health and safety file
	Administer defects liability period
	Sign-off final account

Adapted from Cooke and Williams (2009).

Table 1.6 The planning process of the construction contractor.

Project plan	Action
1. Pre-tender planning	Decision to tender
	Pre-tender arrangements
	Site visit report
	Enquiries to subcontractors and suppliers
	Tender Method Statement
	Build-up estimate
	Pre-tender programme
	Build up preliminaries
	Response to pre-construction Health and Safety information
	Tender risk assessment
	Management adjudication
	Analysis of tender performance
2. Pre-contract planning	Pre-contract meeting and arrangements for commencing work
	Place subcontractor orders
	Site layout planning
	Construction method statement
	Master programme
	Requirement schedules
	Contract budget forecasts
	Risk Assessment
	Preparation and approval of the construction health and safety plan
3. Contract planning	Monthly Planning (long term)
	Weekly Planning (short term)
	Progress Reporting
	Cost and Value Reconciliation
	Report to management
	Review/update the health and safety plan

Adapted from Cooke and Williams (2009).

227

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Levels of Planning

1. Project planning carried out by the client/project manager
2. Pre-tender planning carried out by the tendering contractors
3. Pre-contract planning carried out by the main contractor
4. Contract planning carried out by the main contractor and subcontractors

228

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1. Project planning

starts with the client team or organisation when a programme (or master schedule) will be prepared by the client's agent, representative or project manager.

Reasons for project planning:

- To establish a realistic project master schedule or programme on which to base the appointments of consultants and contractors and commission the occupancy phase of the scheme
- To identify key dates or gateways at critical stages of the project
- To facilitate control of the design and tendering process
- To identify potential risks to progress and avoid possible delays to project completion and revenue generation
- To facilitate the arrangement and draw-down of client cash funding in order to pay for design and other professional services and make interim payments to contractors
- To establish a realistic time period for the construction stage
- To monitor actual progress and take corrective action

229

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2. Pre-tender planning

Prepared by the contractor as an aid to the tendering process. The pre-tender programme will assist the estimator to the price key method-related items in the bills of quantities and also the contract preliminaries which comprise largely time-related costs.

Reasons for pre-tender planning:

- To establish a realistic contract period on which the tender may be based
- To identify construction methods
- To assess method-related items which affect the bid price
- To aid the build-up of contract preliminaries and plant expenditure
- To aid the tendering process

Project plan	Action
1. Pre-tender planning	Decision to tender Pre-tender arrangements Site visit report Enquiries to subcontractors and suppliers Tender Method Statement Build-up estimate Pre-tender programme Build up preliminaries Response to pre-construction Health and Safety information Tender risk assessment Management adjudication Analysis of tender performance

230

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3. Pre-contract planning

takes place during the period between contract award and commencement of work on site.

Reasons for pre-contract planning:

- To provide a broad outline plans or strategy for the project
- To comply with contract conditions
- To establish a construction sequence on which the master programme may be based
- To identify key project dates
- To highlight key information requirements
- To enable the assessment of contract budgets and cumulative value forecasts
- To schedule key dates with respect to key material and subcontractor requirements

Project plan	Action
2. Pre-contract planning	Pre-contract meeting and arrangements for commencing work Place subcontractor orders Site layout planning Construction method statement Master programme Requirement schedules Contract budget forecasts Risk Assessment Preparation and approval of the construction health and safety plan

231

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4. Contract planning

Is done by the main contractor in order to maintain control and ensure that the project is completed on time and within the cost limits established at the tender stage. Subcontractors contribute to the process either by submitting their work programme for approval or through discussion with the main contractor

Reasons for pre-tender planning:

- To monitor the master programme- monthly, weekly and daily
- To plan site operations in detail in the short term
- To optimise and review resources
- To keep the project under review and report no variances

Project plan	Action
3. Contract planning	Monthly Planning (long term) Weekly Planning (short term) Progress Reporting Cost and Value Reconciliation Report to management Review/update the health and safety plan

→ Master Programme

232



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5. Planning a project

- Getting a feel for the project
- Establishing key project dates
- Establishing key activities or events
- Assessing how long the activities will take
- Establishing the sequence
- Deciding which programming technique to use

233

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5. Planning a project

- **Getting a feel for the project**
  - Study the drawing and project documentation
  - Visit the site
  - Assess the scale and scope of the project
  - Assess the approximate value of the project
  - Consider the rate of expenditure (i.e. the relationship between value and time)
- **Establishing key project dates**
  - Project start and finish dates
  - Sectional or phased completion dates
  - Holiday periods
  - Commissioning or handover

234

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5. Planning a project

- Establishing key activities or events
  - Checklist of key activities/events during the design stage
    - Brief the design team
    - Make professional appointments
    - Obtain planning permission
    - Apply for building regulation approval
    - Pre-quality contractors
    - Appoint contractors
    - …
  - Checklist of key activities/events during the tender stage
    - Prepare tender documents
    - Period for tendering
    - Evaluate tenders
    - Compile contract documentation
    - ….
  - Checklist of key activities/events during the pre-contract stage
    - Appoint project staff
    - Prepare requirement schedules
    - Pre-start meetings
- Checklist of key activities/events during the design stage
  - Set up site establishment
  - Groundworks or substructure
  - Frame/external envelope
  - Floors
  - Roof structure and cladding
  - Building watertight
  - Mechanical, electrical and ventilation installations
  - Finishes
  - External works and drainage
  - Practical or substantial completion
  - Clear site
  - …

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6. Video 1

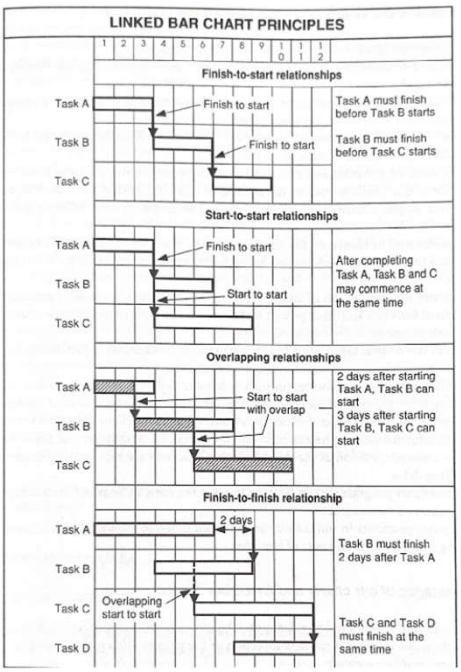


6. Programming techniques

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6.1 Bar chart and Linked Bar Chart

- Finish-to-start relationships
- Start-to-start relationships
- Overlapping relationships
- Finish-to-finish relationships



6. Programming techniques

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FACTORY UNIT — PRE-TENDER PROGRAMME — LINKED BAR CHART

No	OPERATION	Dur
1	Set Up Site	2.0
2	Site Clearance	1.0
3	Piling	2.0
4	Pile Caps / Ground Beams	4.0
5	Drainage	4.0
6	Erect Steel Frame	3.0
7	External Brickwork	5.0
8	Ground Floor Slab	3.0
9	Roof Cladding	2.0
10	Wall Cladding	2.0
11	Services - Internal	4.0
12	External Car Park	4.0

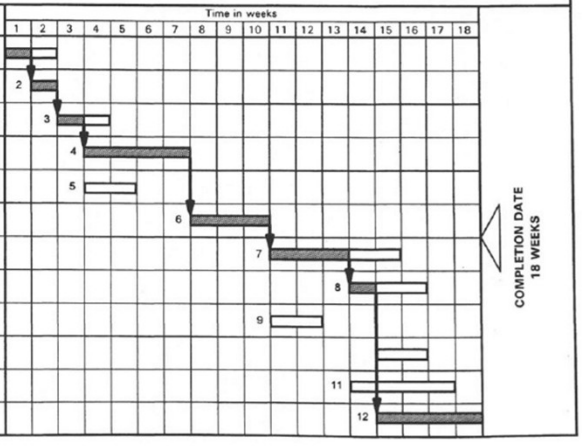


Figure 4.2

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6. Programming techniques

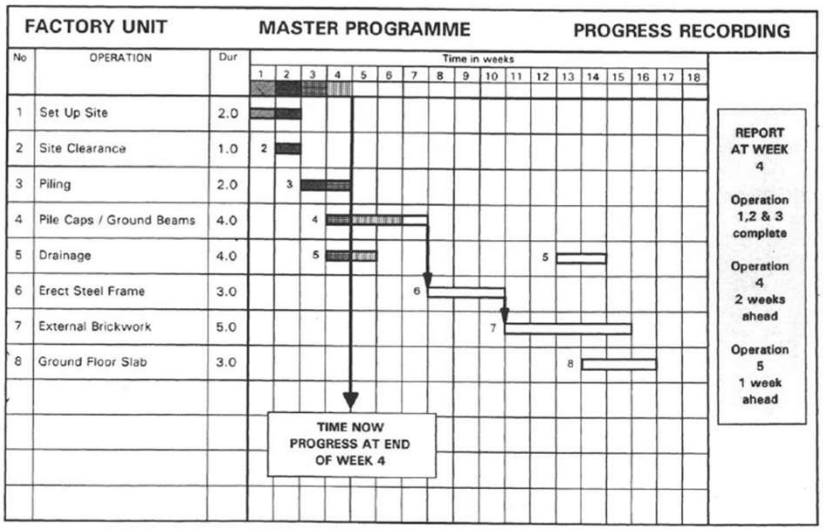
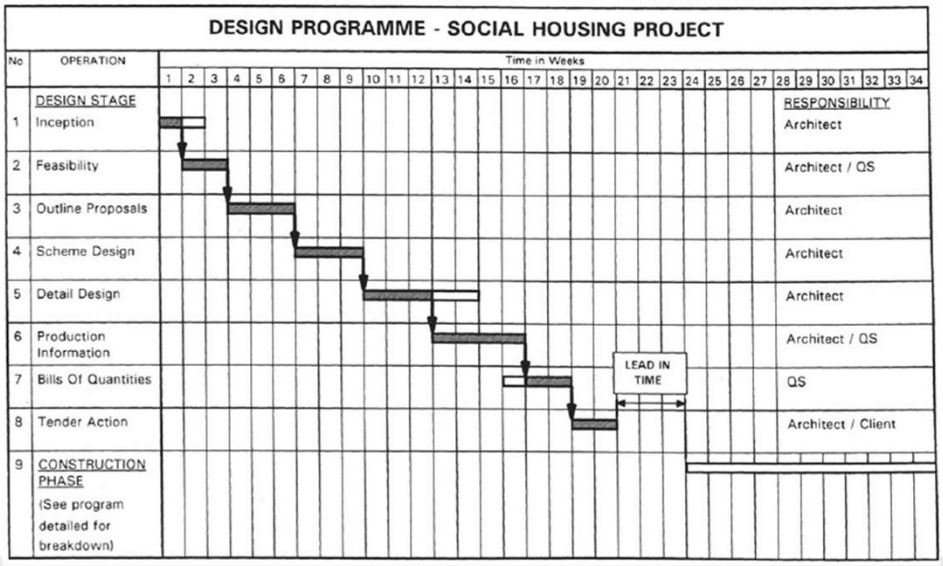


Figure 4.3

239

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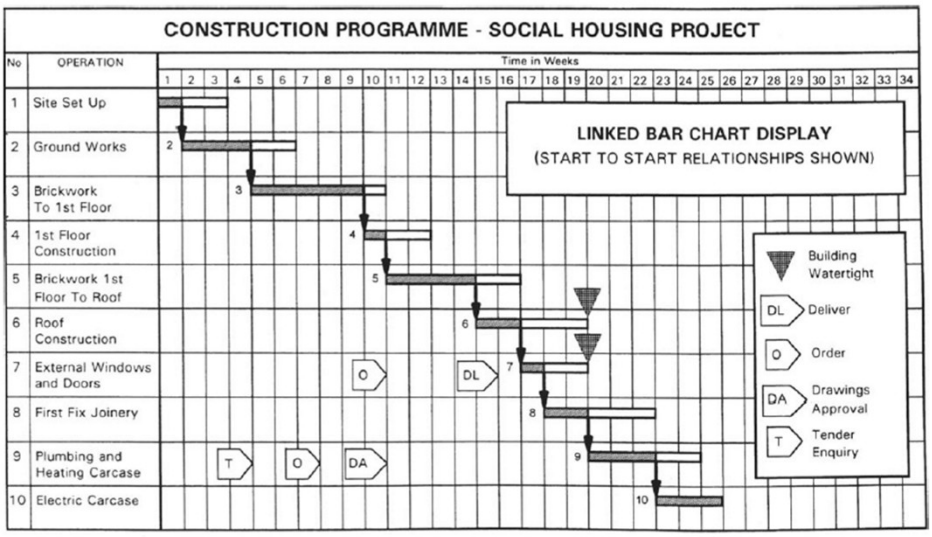
6. Programming techniques



240

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6. Programming techniques



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6. Programming techniques

Disadvantages:

- Does not clearly indicate which operations directly relate to the successful completion of the project. Management by exception therefore cannot be applied to the construction sequence.
- Complex interrelationships cannot be clearly shown.

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6. Programming techniques

6.2 Arrow diagrams (Activity-on-arrow Network)

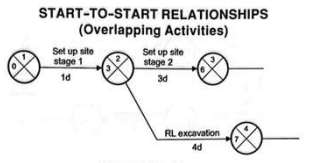
Arrow- activity

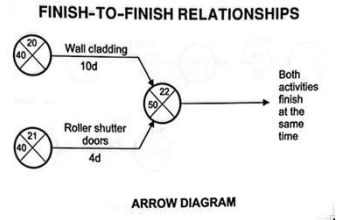
Circle or node – events

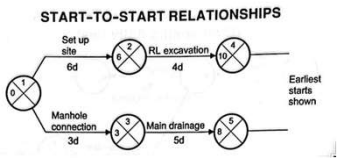
Length of arrow - duration

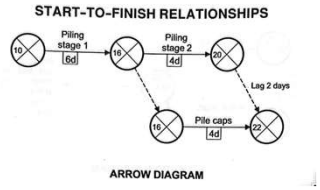
Dummy activities- have no duration or value, to indicate dependencies not shown by the arrow activies

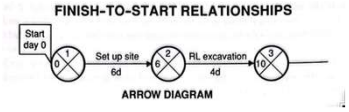
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6. Programming techniques

Total float and critical path

- **Total float (TF): maximum time an activity can be postponed from its early start without delaying the entire project**
  TF = LS- ES or LF-EF
- **Critical path: activities which do not allow for delay (TF = 0)**
- **Free float (FF): delays within the total float that will not affect its successor**

$$FF_i = \min(ES_{i+1}) - EF_i$$

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6. Programming techniques

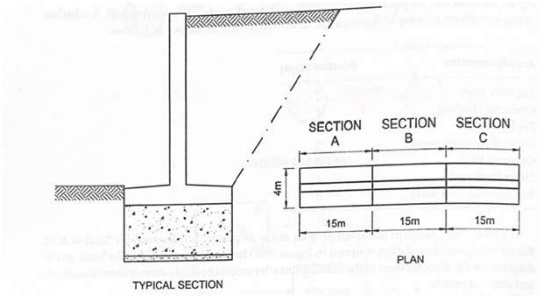
The stages invovled in preparing a network based on arrow diagram principles are as follows:

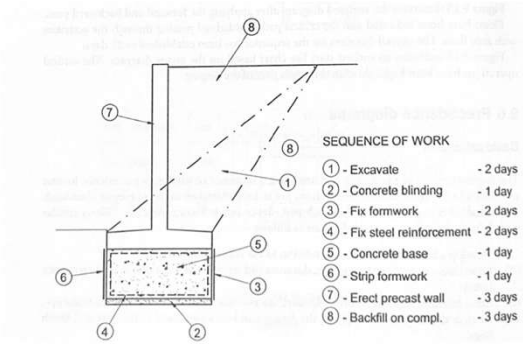
1. Develop a basic logic diagram (doodle diagram) which is based on the sequence of work
2. Enter the operation descriptions, durations and event numbers
3. Apply the forward pass working from left to right through the diagram and calculate the earliest event times for each activity. At an intersection, always carry the highest number forward. Establish the overall project period and enter on the finish flag
4. Apply the backward pass, working from right to left back through the diagram and establish the latest event times for each activity. At intersections always carry the lowest number back.
5. Establish the float times for each activity and highlight the critical path (the activity which have zero float)
6. Convert the arrow diagram to a bar chart format as required, for example an earliest or latest start bar chart display.

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6. Example: Retaining wall project

Figures indicate the plan and section of three bays of a concrete retaining wall. A schedule of the operations relating to the sequence of work has been developed as follows





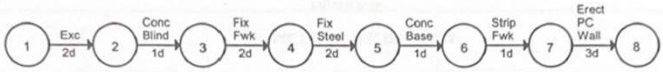
247

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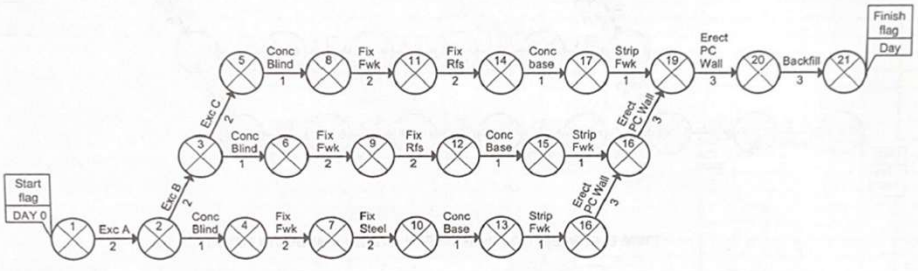
6. Example: Retaining wall project

SEQUENCE OF WORK

①	- Excavate	- 2 days
②	- Concrete blinding	- 1 day
③	- Fix formwork	- 2 days
④	- Fix steel reinforcement	- 2 days
⑤	- Concrete base	- 1 day
⑥	- Strip formwork	- 1 day
⑦	- Erect precast wall	- 3 days
⑧	- Backfill on compl.	- 3 days



LOGIC DIAGRAM FOR BAY A (Section A) FOR RETAINING WALL



INITIAL NETWORK DIAGRAM

Figure 9.9

248

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6. Example: Retaining wall project



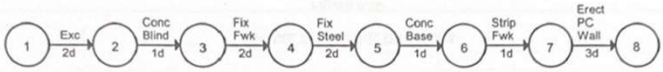
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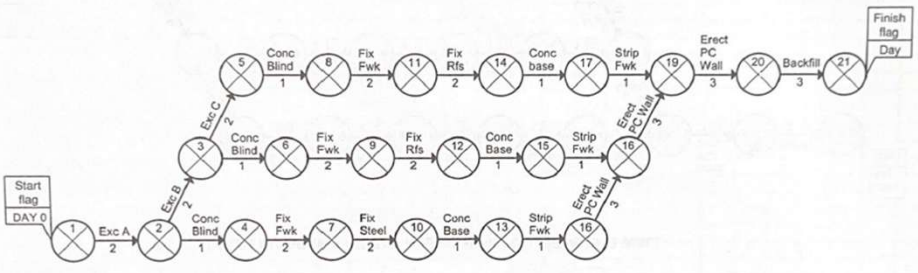
6. Example: Retaining wall project

SEQUENCE OF WORK

(1)	- Excavate	- 2 days
(2)	- Concrete blinding	- 1 day
(3)	- Fix formwork	- 2 days
(4)	- Fix steel reinforcement	- 2 days
(5)	- Concrete base	- 1 day
(6)	- Strip formwork	- 1 day
(7)	- Erect precast wall	- 3 days
(8)	- Backfill on compl.	- 3 days



LOGIC DIAGRAM FOR BAY A (Section A) FOR RETAINING WALL



INITIAL NETWORK DIAGRAM

Figure 9.9

250

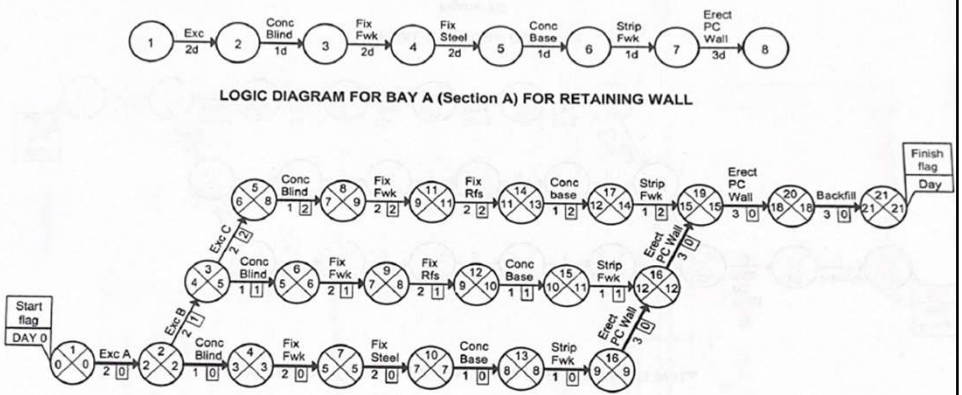
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6. Example: Retaining wall project

3. Total Float:

TF = LS- ES=LF-EF

4. Critical path



LOGIC DIAGRAM FOR BAY A (Section A) FOR RETAINING WALL

INITIAL NETWORK DIAGRAM

Figure 9.10

251

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6. Earliest start bar chart based on the arrow diagram

BAR CHART – BASED ON NETWORK DIAGRAM

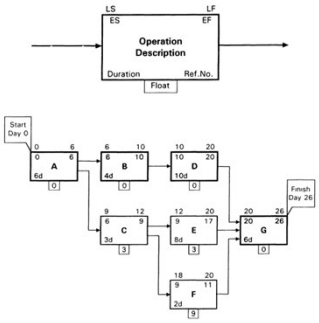
Op No	OPERATIONS	ES	D	FL
	Exc. A	0	2	0
	Blind	2	1	0
	Formwork to base	3	2	0
	Reinforcement	5	2	0
	Concrete	7	1	0
	Strip Formwork	8	1	0
	Erect Precast wall	9	3	0
	Exc. B	2	2	1
	Blind	4	1	1
	Formwork to base	5	2	1
	Reinforcement	7	2	1
	Concrete	9	1	1
	Strip Formwork	10	1	1
	Erect Precast wall	12	3	0
	Exc. C	4	2	2
	Blind	6	1	2
	Formwork to base	7	2	2
	Reinforcement	9	2	2
	Concrete	11	1	2
	Strip Formwork	12	1	2
	Erect Precast wall	15	3	0
	Back fill	18	3	0

252

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6.3 Precedence diagrams (Activity-On-The Node)

- Box or node: activity;
- Linking arrow: the relationship of the activities to one another
- Early start (ES): earliest date on which an activity can start without violating the logic
- Early finish(EF): earliest date on which an activity can finish without violating the logic
- Late Start (LS): latest date on which an activity can start without violating the logic
- Late finish (LF): lastest date on which an activity can finish without violating the logic



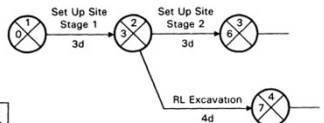
PRECEDENCE NOTATION

253

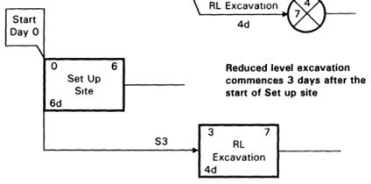
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6. Programming techniques

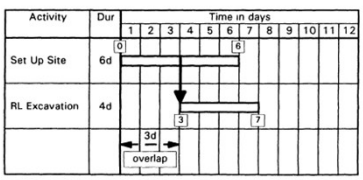
Arrow Diagram



Precedence Box

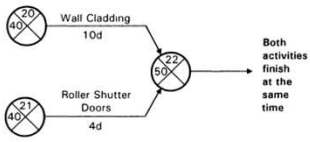


Linked Bar Chart

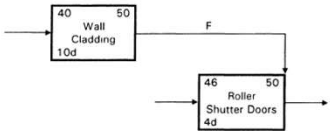


START TO START RELATIONSHIPS
(Overlapping Activities)

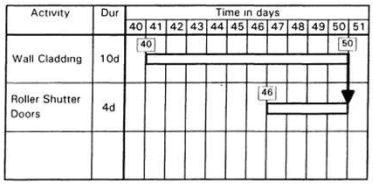
Arrow Diagram



Precedence Box



Linked Bar Chart



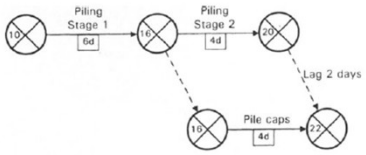
FINISH TO FINISH RELATIONSHIPS

254

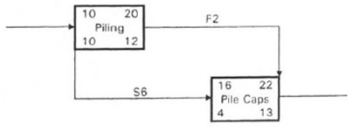
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6. Programming techniques

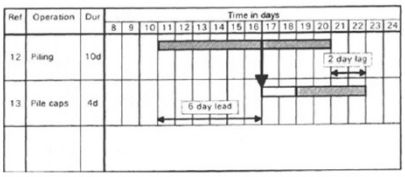
Arrow Diagram



Precedence Box



Linked Bar Chart



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6. Programming techniques

Total float and critical path

- Total float (TF): maximum time an activity can be postponed from its early start without delaying the entire project
  TF = LS- ES or LF-EF
- Critical path: activities which do not allow for delay (TF = 0)
- Free float (FF): delays within the total float that will not affect its successor

$$FF_i = \min(ES_{i+1}) - EF_i$$

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6. Programming techniques

The stages invovled in preparing a network based on arrow diagram principles are as follows:

1. Develop a basic logic diagram in relation to the construction sequence
2. Enter data-operation description, duration and reference box number in the precedence boxes
3. Enter relationships, that is finish-to-start, start-to-start, and start-to-finish relationships.
4. Analyse the diagram by applying the forward and backward pass. Enter start and finish flags
5. Calcuate the floats and establish the operation with zero float –highlight the critical operations. The precedence boxes are much easier to anlyse, as the floats are simply the difference between the umbers in the corner of the precedence boxes.
6. Convert the precedence sequence to a bar chart, if required.

257

7

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PRECEDENCE DIAGRAM

INITIAL PRECEDENCE DIAGRAM

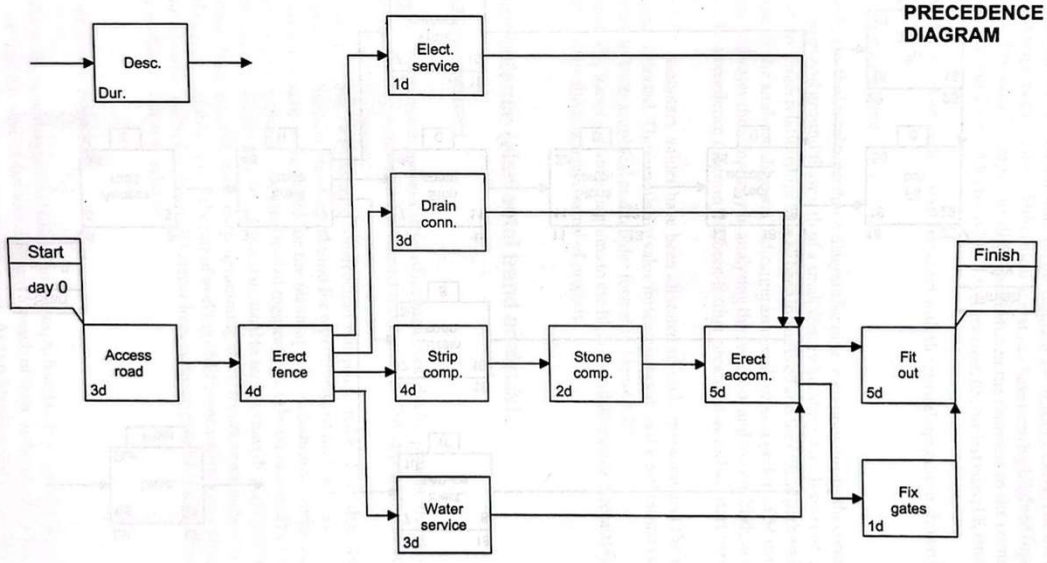
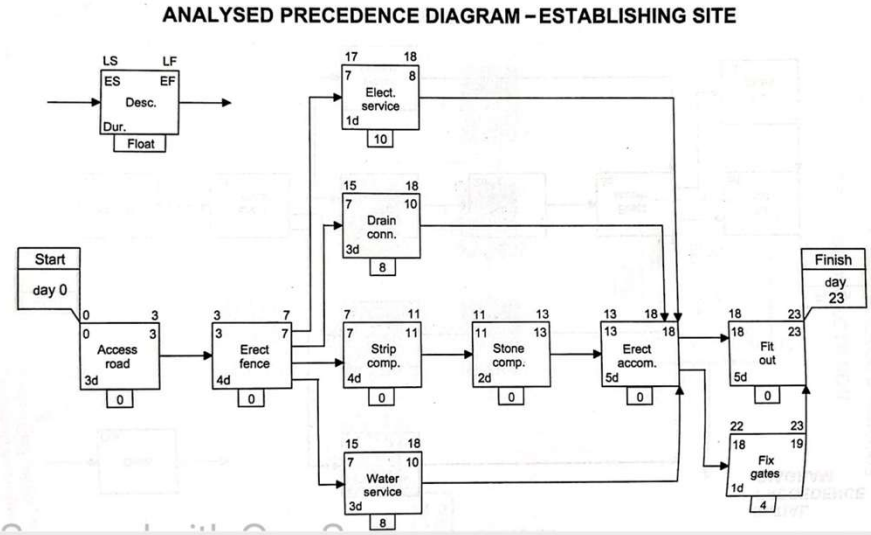


Figure 9.13

258

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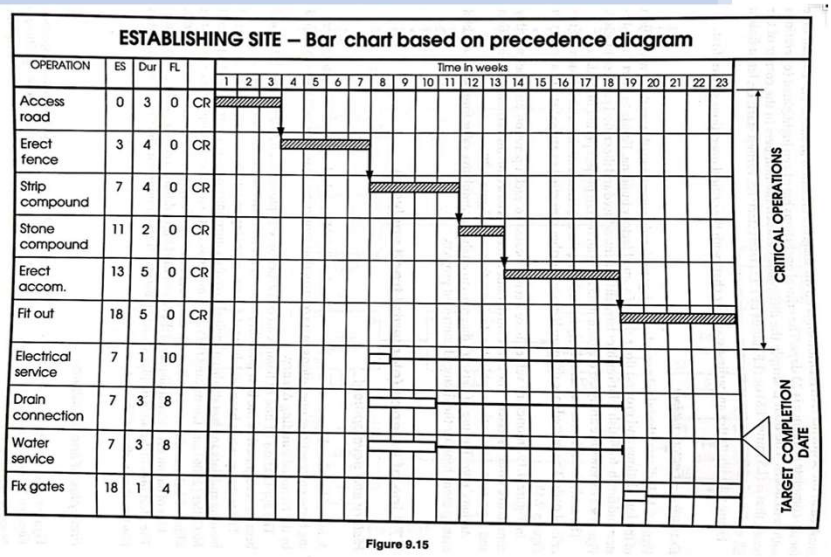
7. Example: Establishing site



259

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7. Example: Establishing site



OPERATION	ES	Dur	FL	
Access road	0	3	0	CR
Erect fence	3	4	0	CR
Strip compound	7	4	0	CR
Stone compound	11	2	0	CR
Erect accom.	13	5	0	CR
Fit out	18	5	0	CR
Electrical service	7	1	10	
Drain connection	7	3	8	
Water service	7	3	8	
Fix gates	18	1	4	

Figure 9.15

260

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8. Example: Factory project

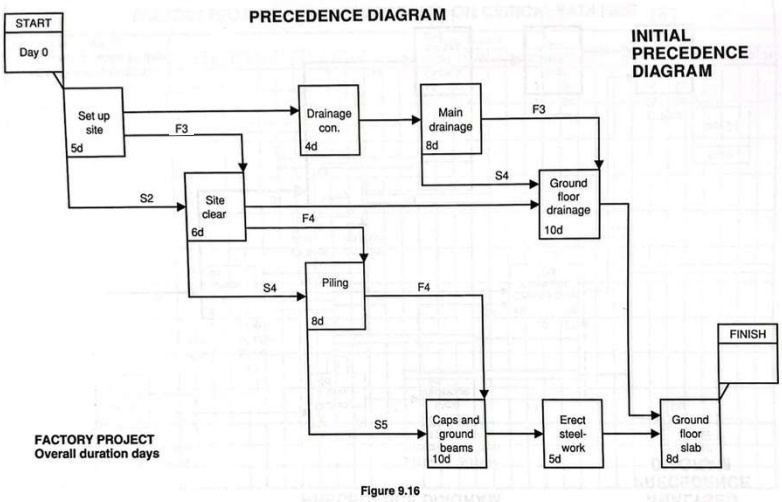


Figure 9.16

261

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8. Example: Factory project

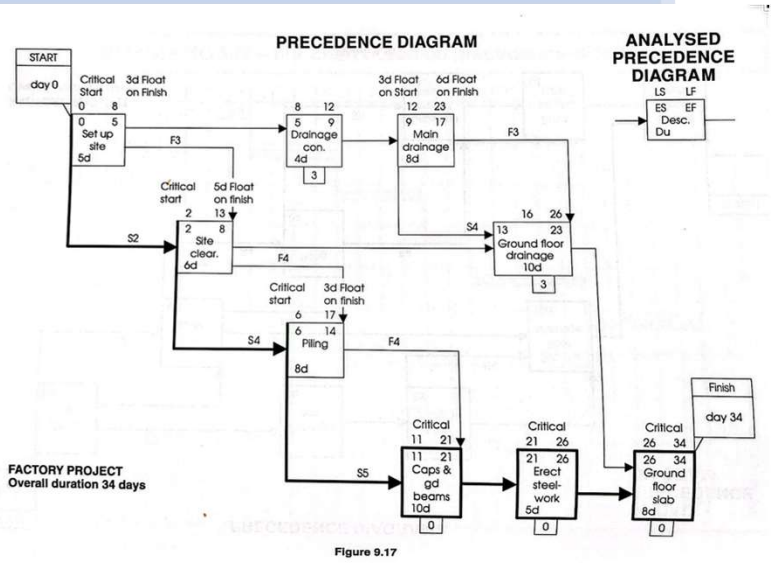


Figure 9.17

262

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8. Example: Factory project

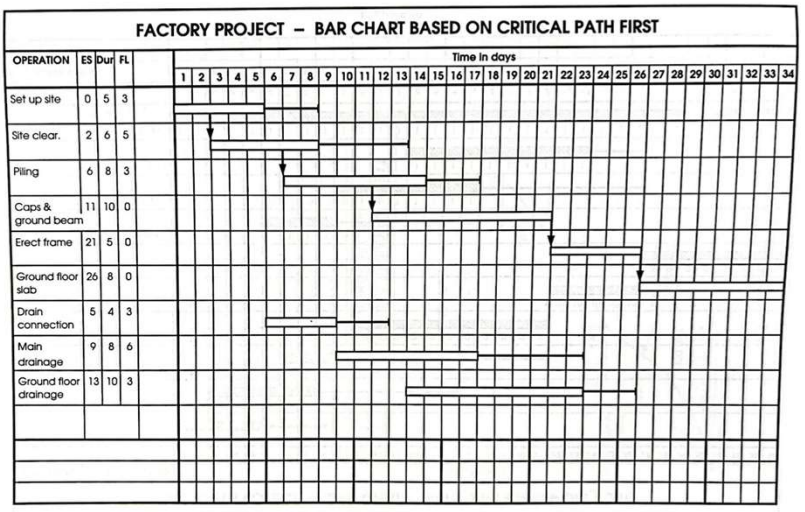


Figure 9.18

263

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8. Example: Factory project

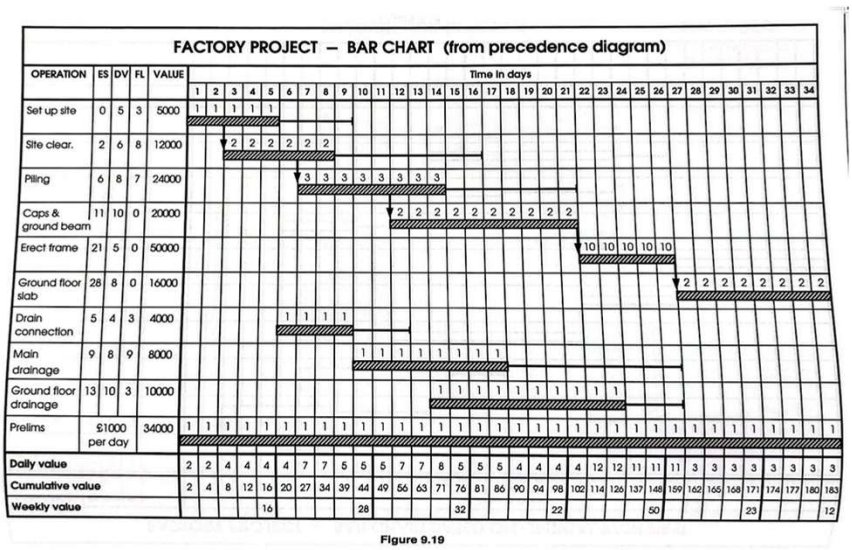


Figure 9.19

264